

Counting tournaments with domination number at least three: a proved polynomial-time algorithm

Computational derivation and validation

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1. Introduction and main theorem

A *tournament* on n vertices is an orientation of the complete graph K_n : for every pair $\{u, v\}$ exactly one of $u \rightarrow v$, $v \rightarrow u$ holds. Its *domination number* $\gamma(T)$ is the size of the smallest set D of vertices such that every other vertex is beaten by some member of D . For $n \geq 2$ we give and prove correct an algorithm that computes

$$L(n) = \#\{\text{labeled tournaments on } [n] \text{ with } \gamma(T) \geq 3\}.$$

(We take $n \geq 2$ throughout: for $n = 1$ the reduction below would return 1 while $L(1) = 0$.) The algorithm runs in **polynomial time**: $L(n)$ is a signed sum over caterpillar forests and odd-cycle suns whose per-graph weight is computed by a linear-time transfer recurrence (§§3–5), and a symmetric-power / exponential-generating-function aggregation (§6) evaluates the entire signed sum *without enumerating the graphs*, in $\text{poly}(n)$ big-integer operations – as against the $2^{\binom{n}{2}} = 2^{\Theta(n^2)}$ of direct evaluation. (Summing the same per-graph weights by naïve term-by-term enumeration instead gives a $2^{O(n)}$ variant; we mention it only as a special case and do not rely on it.)

Correctness. Each step is a theorem: the master reduction and sparsity restriction (§2), the checkerboard factorization and odd-cycle-sun count (§3), the per-caterpillar transfer (§4), and the polynomial aggregation (**Theorem L3**, §5); so $L(n)$ is computed exactly in polynomial time. Where an independent check is available it agrees – $L(7), L(8), L(9)$ match direct brute-force enumeration, the isomorphism counts through $n = 10$ match McKay’s catalogues, and $L(11)$ satisfies a Burnside congruence (§6) – but these are corroboration, not the proof.

The tractability here should not be mistaken for a complexity-theoretic breakthrough: we *count* rather than *decide*, and the two are unrelated. The count sums over the sparse family supplied by the structure theorem of Fisher, Lundgren, Merz and Reid (§2) – which is where essentially all the tractability comes from – so no individual instance is ever solved. (Deciding $\gamma(T) \leq k$ for a *given* tournament is a genuinely harder-looking problem – quasi-polynomial and LOGSNP-complete: Megiddo–Vishkin, *Theoret. Comput. Sci.* **61** (1988) 307–316; Papadimitriou–Yannakakis, *J. Comput. System Sci.* **53** (1996) 161–170 – but that hardness does not bear on the count, and no barrier is at issue.) A Burnside/cycle-index step yields the count of $\gamma \geq 3$ tournaments *up to isomorphism* (a separate small- n layer, §6), extending a sequence previously tabulated only to $n = 10$; we are aware of no prior formula, recurrence, or OEIS entry for it, so the values for $n \geq 11$ are new.

The condition $\gamma \geq 3$ is the two-paradox (Schütte) property S_2 : no single vertex beats everyone, and no pair of vertices together beats all others. By Reid, McRae, Hedetniemi and Hedetniemi (Australas. J. Combin. **29** (2004) 157–172, Thm. 5), $\gamma(T) \leq 3$ for every tournament on $n < 19$ vertices, so for $2 \leq n \leq 18$ the count “ $\gamma \geq 3$ ” equals “ $\gamma = 3$ exactly,” and the domination distribution is supported on $\{1, 2, 3\}$ with $\#\{\gamma = 1\} = n 2^{\binom{n-1}{2}}$.

Scope. The method rests on the structure theorem for dominating *pairs* – $\gamma \geq 3$ is exactly “dom(T) edgeless” – so it computes $\#\{\gamma \geq 3\}$ and nothing finer; there is no analogous characterization for dominating triples, and $\gamma \geq 4$ is out of reach. For $n \leq 18$ this costs nothing: $L(n) = \#\{\gamma = 3\}$, the statistic of interest, and even the single-exponential form of the method reaches that whole range. At $n = 19$ the identity breaks – the Paley tournament QRT_{19} has $\gamma = 4$ – so for $n \geq 19$, $L(n) = \#\{\gamma \geq 3\}$ is a blend of $\gamma = 3$ and $\gamma \geq 4$ that the method cannot separate. Beyond $n = 18$ the polynomial algorithm therefore evaluates this aggregate count efficiently; it does not recover $\#\{\gamma = 3\}$.

2. Inclusion–exclusion and structural reduction

For a vertex w write $N^+(w)$ for its *out-neighborhood* (the set of vertices w beats). A pair $\{u, v\}$ is a **dominating pair** if no vertex beats both, i.e. $\{u, v\}$ is not contained in any $N^+(w)$. The **domination graph** $\text{dom}(T)$ has an edge $\{u, v\}$ exactly for the dominating pairs. Then

$$\gamma(T) \geq 3 \iff \text{no vertex dominates and no pair dominates} \iff \text{dom}(T) \text{ has no edge.}$$

Indeed $\gamma = 1$ means some vertex beats all others, and $\gamma \leq 2$ that some pair $\{u, v\}$ beats everyone else – exactly that $\{u, v\}$ has no common in-dominator, i.e. is an edge of $\text{dom}(T)$.

Equivalently, $\{u, v\}$ is a dominating pair iff no out-neighborhood contains both u and v , i.e. every $N^+(w)$ is an independent set in the graph having the single edge $\{u, v\}$. Asking a *set* of pairs H to all be dominating is therefore

$$H \subseteq \text{dom}(T) \iff \text{every out-neighborhood } N^+(w) \text{ is } H\text{-independent.}$$

The master inclusion–exclusion. Index tournaments by their set of dominating pairs and Möbius-invert on the (edge-)lattice of graphs on $[n]$: since $L(n) = \#\{T : \text{dom}(T) = \emptyset\}$,

$$L(n) = \sum_{H \subseteq K_n} (-1)^{e(H)} N(H), \quad N(H) := \#\{T : H \subseteq \text{dom}(T)\},$$

where $e(H)$ is the number of edges of H . Now $N(H)$ depends on H only through its non-isolated part. Let H have exactly v non-isolated vertices carrying a graph H' (no isolated vertices) and $n - v$ untouched vertices. The constraint “every $N^+(w)$ is H -independent” involves only how each vertex relates to the v support vertices, so the tournament factorizes:

- the sub-tournament on the $n - v$ non-support vertices is **arbitrary**: $2^{\binom{n-v}{2}}$;
- the sub-tournament on the v support vertices must have **every out-neighborhood H' -independent**: call the number of such sub-tournaments $T(H')$;

- each non-support vertex w may beat any H' -independent subset of the support (and is beaten by the rest): $I(H')$ choices each, where $I(H')$ is the number of independent sets of H' (all sizes, including the empty set). This contributes $I(H')^{n-v}$.

Summing over the choice of support ($\binom{n}{v}$ ways) and over H' gives the

Master formula.

$$\boxed{L(n) = \sum_{v=0}^n \binom{n}{v} 2^{\binom{n-v}{2}} A_v(n-v)}, \quad A_v(t) = \sum_{\substack{H \text{ on } [v] \\ \text{no isolated vertex}}} (-1)^{e(H)} T(H) I(H)^t,$$

the inner sum ranging over *labeled* graphs H on v vertices with minimum degree ≥ 1 , and

$$T(H) = \#\{\text{tournaments on } V(H) : \text{every out-neighborhood is } H\text{-independent}\}.$$

Here $A_0 = 1$ (empty graph) and $A_1 = 0$ (a single vertex is isolated). The $v = 0$ term is $2^{\binom{n}{2}}$, the total number of tournaments, and the higher terms are the inclusion–exclusion corrections subtracting the $\gamma \leq 2$ tournaments.

Everything now rests on computing $T(H)$ and enumerating the H that matter.

Sparsity (Fisher–Lundgren–Merz–Reid). $T(H) > 0$ requires $H \subseteq \text{dom}(T)$ for some tournament T , and the structure of domination graphs is completely known:

Theorem (Fisher, Lundgren, Merz, Reid, *J. Graph Theory* 29(2) (1998) 103–110). For every tournament T , the domination graph $\text{dom}(T)$ is either a forest of caterpillars, or an odd cycle together with pendant and isolated vertices.

The master formula strips isolated vertices before forming H' , so the relevant consequence is that the non-isolated part of any domination graph is either a forest of nontrivial caterpillars or a single **odd-cycle sun** (an odd cycle C_g , $g \geq 3$, with any number of pendant leaves on its vertices). (A *caterpillar* is a tree that becomes a path when its leaves are deleted; a spine path $s_0 \cdots s_{m-1}$ with $\ell_i \geq 0$ leaves on s_i .) Consequently $T(H') = 0$ unless every component of H' is a nontrivial caterpillar, or H' is a single odd-cycle sun. This restricts the sum from all $2^{\binom{v}{2}}$ graphs to caterpillar forests and suns, but is **not** by itself a polynomial enumeration: there are $\Theta(2^k)$ caterpillars on k vertices (OEIS A005418), so the structural theorem alone gives only a single-exponential ($2^{\Theta(n)}$) enumeration, not $\text{poly}(n)$.

3. Weights of connected components

Theorem. Let H be a forest, **each of whose components** $C_1, \dots, C_c$ **has at least one edge** (as always holds for the isolate-free support H'). Then

$$T(H) = 2^{\binom{c}{2}} \prod_{i=1}^c \text{Within}(C_i), \quad \text{Within}(C) := T(C).$$

Under the same hypothesis (no isolated vertices), if some component is cyclic then $T(H) = 0$ unless H is a single odd-cycle sun, in which case $\text{Within} = 2^{1+\sum_i \binom{\ell_i}{2}}$ (in particular $\text{Within} = 2$ when

no cycle vertex carries two or more leaves); an even cycle gives 0. (The no-isolates hypothesis is needed here too – e.g. $T(C_3 \sqcup K_1) = 8 \neq 0$ – for the reason given next.)

The edge-bearing hypothesis is essential: for $H = K_2 \sqcup K_1$ (an isolated vertex) one has $T = 2 \cdot 3 = 6$ (the isolated vertex may beat at most one endpoint of the edge), whereas the displayed formula would give $2^{\binom{2}{2}} \cdot 2 \cdot 1 = 4$. Isolated vertices are exactly what the master formula removes, so no such H occurs there; but the theorem must exclude them.

Proof. A set S is H -independent iff its restriction to each component is component-independent (all edges live inside components). Fix the within-component orientations. Two facts remain:

- (i) *Within factor.* The constraint “ $N^+(w) \cap C_i$ is C_i -independent” for $w \in C_i$ depends only on the sub-tournament on C_i ; these constraints are independent across components, giving $\prod_i \text{Within}(C_i)$.
- (ii) *Between-component factor (checkerboard lemma).* Fix an ordered pair (C_i, C_j) and write $X_{uv} = 1$ iff $u \in C_i$ beats $v \in C_j$. For any edges $uu' \in E(C_i)$ and $vv' \in E(C_j)$ consider

$$\begin{pmatrix} X_{uv} & X_{uv'} \\ X_{u'v} & X_{u'v'} \end{pmatrix}.$$

Independence of the out-neighbourhoods in C_j forces neither row to be $(1, 1)$ (else some u beats the adjacent v, v'); independence of the reverse out-neighbourhoods in C_i forces neither column to be $(0, 0)$ (else some v beats the adjacent u, u'). A 2×2 0/1 matrix with no all-ones row and no all-zeros column has exactly two ones and is one of $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Since each C_i, C_j is a *connected bipartite* graph (a caterpillar has ≥ 1 edge), this local constraint propagates: writing p_i, p_j for the bipartition indicators, every valid cross-orientation has the form $X_{uv} = b \oplus p_i(u) \oplus p_j(v)$ for a single bit $b \in \{0, 1\}$. Hence exactly **two** cross-orientations per component pair (connectedness and the edge in each component are what pin it to 2; a bare K_1 has no edge to propagate b and would allow 3, the source of the K_1 discrepancy above). Over the $\binom{c}{2}$ pairs this gives $2^{\binom{c}{2}}$. Cyclic components are killed unless the odd-cycle-sun exception applies. $\square$

We have verified the theorem exhaustively for every isolate-free graph on $v \leq 9$ vertices (zero discrepancies). The “2 per component pair” mechanism is illustrated by $T(K_2 \sqcup K_2) = 2^1 \cdot 2 \cdot 2 = 8$, whose 8 tournaments have exactly 2 valid cross-orientations for each of the 2×2 within-orientations.

Lemma L1 (odd-cycle sun). $\text{Within}(\text{odd-cycle sun}) = 2^{1+\sum_i \binom{\ell_i}{2}}$; $\text{Within}(C_{2m}) = 0$; and any connected graph containing a cycle that is *not* an odd-cycle sun has $\text{Within} = 0$.

Zero cases. If $\text{Within}(H) = T(H) > 0$ then $H \subseteq \text{dom}(T)$ for some T ; by the Fisher–Lundgren–Merz–Reid theorem (§2) a connected cyclic contributing graph must be a spanning connected cyclic subgraph of an odd-cycle sun, hence is itself an odd-cycle sun. So any connected cyclic H that is not an odd-cycle sun has $\text{Within} = 0$; in particular an even cycle gives 0.

The value. Write the cycle as $c_0, \dots, c_{g-1}$, with ℓ_i leaves on hub c_i . Working inside any valid tournament (using only the dominating-pair condition on cycle edges and H -independence of out-neighbourhoods), four facts hold: **(a)** the cycle is one of exactly two directed rotations and every chord $c_i \rightarrow c_{i+d}$ is forced (d odd outward), consistent only for odd g (even g forces the wrap edge both ways, giving 0); **(b)** each leaf’s orientation against every foreign hub is forced by the parity of the offset; **(c)** every foreign leaf–leaf edge is forced; **(d)** only the $\binom{\ell_i}{2}$ edges between same-hub leaves are free, and every such completion is valid. Each of the two rotations therefore contributes

$\prod_i 2^{\binom{\ell_i}{2}}$, giving $\text{Within} = 2^{1+\sum_i \binom{\ell_i}{2}}$. The four claims are established by a parity induction on the cycle-and-chord orientations, given in Appendix B. $\square$

Thus the problem reduces to the single primitive $\text{Within}(\text{caterpillar})$, computed in §4. The smallest values – star $K_{1,m} = 2^{\binom{m}{2}} + m 2^{\binom{m-1}{2}}$, path $P_k = 4$ ($k \geq 3$), $K_2 = 2$ as a component, and the double-/triple-star channels – are closed forms (Appendix A), all brute-verified (§6).

4. The caterpillar transfer theorem

This section gives the per-caterpillar primitive: it computes $\text{Within}(C)$ in time linear in $|C|$ by an explicit 4-dimensional spine recurrence, proved exact for all caterpillars (the transfer theorem below). A caterpillar is encoded by its leaf-count word $\mathbf{l} = (\ell_0, \dots, \ell_{m-1})$.

Leaf-twin symmetry. All ℓ_i leaves on s_i are false twins, so validity depends only on how *many* of them each out-neighborhood contains. Writing

$$p_A(\ell) = 2^{\binom{\ell}{2}} A^\ell \quad \text{and} \quad q_A(\ell) = \ell 2^{\binom{\ell-1}{2}} A^{\ell-1}, \quad A \in \{1, 2, 4\},$$

the per-vertex leaf factor is a combination of these two shapes, where A is the (finite, per-core) number of admissible one-leaf extensions (the a-priori slot $A = 8$ never occurs). The 4-dimensional state is **not** motivated by counting these scalar A -values; it comes from the boundary profile $\pi(P) = (\text{Within}(P \cdot [0]), \dots, \text{Within}(P \cdot [3]))$ – four evaluations whose governing matrix G is nonsingular ($\det G = 144$), as proved in the transfer theorem below.

Leaf-Factor Lemma (proved). Fix all edges not incident to the leaves of one spine vertex s (an internally-valid core config τ). Summing over the leaf-orientations of s contributes exactly $f_\tau(\ell) = p_A(\ell) + A^* q_A(\ell)$, where $A = \#\{\text{C-independent } Q \subseteq W \text{ containing the external in-neighbours of } s\}$ and A^* is the same count restricted to sets avoiding s 's C-neighbours. (Proof: at most one leaf beats s ; the two cases give the two terms.) This is the true origin of the two-term basis.

Channels. The star, double-star and triple channels are *theorems* for every leaf-count ℓ , obtained from the Leaf-Factor Lemma by a finite enumeration of internally-valid core configurations: each channel is a $\mathbb{Z}$ -combination of the two-dimensional block $\{p_A, q_A\}$, and enumerating the finitely many cores fixes the two coefficients. The single-vertex value is $\text{star}(m) = 2^{\binom{m}{2}} + m 2^{\binom{m-1}{2}} = p_1(m) + q_1(m)$ by direct counting. The fourteen double-/triple-channel cases, their core-config $(A:\text{count})/(A:\sum A^*)$ table, and the exhaustiveness argument are collected in Appendix A, together with the resulting closed forms $DS(L, d)$ and $g_{a,c}(\ell)$.

State and recurrence. Realize the automaton in the basis of *one-symbol suffixes*: for a processed spine prefix P , keep the 4-vector

$$R[j] = \text{Within}(P \text{ followed by a spine vertex carrying } j \text{ leaves}), \quad j \in \{0, 1, 2, 3\}.$$

Let G be the fixed 4×4 matrix $G[a][d] = DS(a, d) = \text{Within}([a, d])$ (double-star values), with $\det G = 144$ and adjugate $\text{adj}(G) = 144 G^{-1}$ (an explicit integer matrix). Let $W(\ell)$ be the 4×4 matrix $W(\ell)[j][a] = \text{Within}([a, \ell, j])$ (the triple values, each a fixed combination of the two shapes above). Then:

- **Initialize** $R^{(0)}[j] = \text{Within}([\ell_0, j])$ from the double-star channel.

- **Advance** across each interior symbol ℓ : $R' = W(\ell) G^{-1} R$.
- **Read out** with the last symbol $x = \ell_{m-1}$: $\text{Within}(\mathbf{1}) = \lambda(x)^\top R$, $\lambda(x) = G^{-1}w(x)$, $w(x)[a] = \text{Within}([a, x])$.

Compactly,

$$\text{Within}(\ell_0, \dots, \ell_{m-1}) = \lambda(\ell_{m-1})^\top \left(\prod_{i=m-2}^1 W(\ell_i) G^{-1} \right) R^{(0)}(\ell_0).$$

Here $G[a][d] = DS(a, d)$ has $\det G = 144$, and $\text{adj}(G) = 144 G^{-1}$ is the integral matrix

$$\text{adj}(G) = \begin{pmatrix} 0 & -48 & 24 & 0 \\ -48 & -30112 & 22928 & -2088 \\ 24 & 22928 & -17416 & 1584 \\ 0 & -2088 & 1584 & -144 \end{pmatrix} \quad (\text{adj}(G)G = 144I, \text{ elementary}).$$

Each spine vertex is a constant number of 4×4 big-integer operations, so $\text{Within}(C)$ costs $O(|C|)$ operations (integers of $O(|C|^2)$ bits) – versus $2^{\binom{|C|}{2}}$ for direct enumeration. (The exact-integer division by 144 at each *interior* step is justified by Theorem L2 below: every interior numerator $W(\ell) \text{adj}(G)R$ is again a vector of tournament counts, hence divisible by 144, and the recurrence runs entirely in exact integers.)

Theorem L2 (caterpillar transfer). For every prefix P the four-coordinate boundary profile $\pi(P) = (\text{Within}(P \cdot [d]))_{d=0}^3$ is a sufficient statistic: it evolves by a prefix-independent matrix, $\pi(P \cdot [\sigma]) = M(\sigma) \pi(P)$ (with $M(\sigma)$ built from the triple values $\text{Within}([a, \sigma, d])$), and the terminal value $x \mapsto \text{Within}(P \cdot [x])$ is the fixed linear readout $\lambda(x)^\top \pi(P)$ with $\lambda(x) = G^{-1}(DS(a, x))_a$. Hence the 4-dimensional recurrence computes $\text{Within}(C)$ exactly for **every** caterpillar.

Proof. Split $C = P \cdot v$ at the cut spine edge $\{s, g\}$, where s is the last spine vertex of P (vertex set $B := V(P)$) and g the first spine vertex of the right part v (vertex set R). Since leaves stay in the block of their hub and the spine is a path, $\{s, g\}$ is the **only** cross C -edge, so a tournament decomposes uniquely as (T_B, T_R, D) with $T_B = T|_B$, $T_R = T|_R$, and cross-data $D_z := N_T^+(z) \cap B \subseteq B$ for $z \in R$.

Lemma 4 (gluing in coordinates). $T = (T_B, T_R, D)$ is valid iff **(0')** T_B, T_R are valid; **(A')** each D_z is C -independent; **(B')** for every C -edge $\{z, z'\}$ of R , $D_z \cup D_{z'} = B$; together with the boundary conditions **(C1)** $I \subseteq D_g$, where $I := N_{T_B}^-(s)$, and **(C2)** $s \notin D_x$ for every $x \in R$ that beats g in T_R . *Proof.* By the Reformulation, T is valid iff every $N^+(w)$ is C -independent, and (as $\{s, g\}$ is the unique cross edge) a set is C -independent iff its B - and R -parts are and it omits $\{s, g\}$. For $w \in R$, $N^+(w) \cap B = D_w$, giving (A'); for $w \in B$, $N^+(w) \cap R = \{z : w \notin D_z\}$ is C -independent iff no C -edge $\{z, z'\}$ of R has $w \notin D_z \cup D_{z'}$, i.e. $D_z \cup D_{z'} = B$ after ranging over w – this is (B'). The requirement that no $N^+(x)$ contain $\{s, g\}$ gives (C1) for $x \in B$ and (C2) for $x \in R$. $\square$

Lemma 5 (covering $\Rightarrow$ colour class). Let B be connected with $|B| \geq 2$ and 2-colouring $\{X, Y\}$. If $D, D' \subseteq B$ are C -independent with $D \cup D' = B$, then $\{D, D'\} = \{X, Y\}$. *Proof.* Any $v \in D \cap D'$ has every C -neighbour u in $D \cup D' = B$; if $u \in D$ then $\{u, v\} \subseteq D$ is a C -edge inside an independent set, and $u \in D'$ is likewise impossible, so v has no C -neighbour – contradicting connectedness with $|B| \geq 2$. Hence $D \cap D' = \emptyset$, so (D, D') is a proper 2-colouring of B ; by uniqueness of the 2-colouring of a connected bipartite graph, $\{D, D'\} = \{X, Y\}$. $\square$

Lemma 6 (two-colouring collapse). For $B = V(P)$ connected with $|B| \geq 2$ and R a connected caterpillar with ≥ 1 edge, the families D satisfying (A')+(B') are exactly the **two** proper

2-colourings of R by the classes $\{X, Y\}$ (a colouring and its swap). *Proof.* Every $z \in R$ meets a C -edge $\{z, z'\}$ of R ; (A')+(B') with Lemma 5 give $\{D_z, D_{z'}\} = \{X, Y\}$, so $z \mapsto D_z$ is a proper 2-colouring of R , and conversely every such colouring satisfies (A')+(B'). A connected graph has exactly two. $\square$

So the cross-data has exactly **two** possibilities, *independent of* $|V(P)|$: the feared “ $i(P)$ blow-up” (a growing independent-set count on P) never appears, because the past vertices’ own out-neighbourhood constraints (B') force each D_z to be an entire colour class. **The two colour classes X, Y are the two boundary states the transfer must carry.**

Summation (L2_proof §§4–5). Name X the colour class containing s . For $R = [\sigma, d]$ (hubs g, h' carrying σ, d leaves) the two colourings $\kappa^\pm$ assign X, Y to the class $R_0 = \{g\} \cup L_{h'}$. Evaluating (C1),(C2) for a fixed valid T_B (fixing $I = N_{T_B}^-(s)$) and a fixed valid T_R : κ^+ needs $I \subseteq X$ and no leaf of h' beats g ; κ^- needs $I \subseteq Y$ and g beats h' with no leaf of g beating g . Writing $N^+(\sigma, d), N^-(\sigma, d)$ for the resulting two junction constants (functions of (σ, d) only) and summing over valid T_R ,

$$\sum_{T_R} \#\{\text{valid } D\} = [I \subseteq X] N^+(\sigma, d) + [I \subseteq Y] N^-(\sigma, d). \quad (4.1)$$

Summing (4.1) over all valid T_B on P (valid because $|V(P)| \geq 2$, so Lemma 6 applies) gives the explicit **rank-2 relation**

$$\text{Within}(P \cdot [\sigma, d]) = a(P) N^+(\sigma, d) + b(P) N^-(\sigma, d), \quad (5.1)$$

$a(P) := \#\{T_B : I \subseteq X\}$, $b(P) := \#\{T_B : I \subseteq Y\}$. The *same* machinery on the one-hub part $R = [d]$ gives $N_{[d]}^+ = \text{star}(d)$ and $N_{[d]}^- = p(d) = 2^{\binom{d}{2}}$, so $\text{Within}(P \cdot [d]) = a(P) \text{star}(d) + b(P) p(d)$; taking $d = 1, 3$ with $\text{star}(1) = 2$, $p(1) = 1$, $\text{star}(3) = 14$, $p(3) = 8$,

$$\text{Within}(P \cdot [1]) = 2a(P) + b(P), \quad \text{Within}(P \cdot [3]) = 14a(P) + 8b(P),$$

so the 2×2 system has determinant 2, uniquely fixing $a(P), b(P)$ as prefix-independent linear functionals of $\pi(P)$. Substituting these back into (5.1):

Proposition 7 (rank-2 transfer). For every prefix P with $|V(P)| \geq 2$ and all $\sigma, d \geq 0$,

$$\text{Within}(P \cdot [\sigma, d]) = c_1(\sigma, d) \text{Within}(P \cdot [1]) + c_3(\sigma, d) \text{Within}(P \cdot [3]), \quad (5.4)$$

with c_1, c_3 prefix-independent (functions of (σ, d) only). Here the two colour classes X, Y are the two boundary quantities, and $\text{Within}(P \cdot [1]), \text{Within}(P \cdot [3])$ are the two basis profiles into which they are read. $\square$

Proposition 8 (profiles of ≥ 2 -vertex prefixes lie in a fixed 3-space). For every prefix P with $|V(P)| \geq 2$, $\pi(P) \in U_0 := \text{span}\{p_1, q_1, \delta_0\}$ (restricted to $\{0, 1, 2, 3\}$), and $\{\pi([1]), \pi([2]), \pi([3])\}$ is a basis of U_0 . *Proof.* This follows directly from the one-hub gluing relation already obtained above – $\text{Within}(P \cdot [d]) = a(P) \text{star}(d) + b(P) p_1(d)$ for every $d \geq 1$ (the $R = [d]$ case of (5.1), valid because $|V(P)| \geq 2$) – with no appeal to a power-of-two lemma. Thus on $x \in \{1, 2, 3\}$ the map $x \mapsto \text{Within}(P \cdot [x])$ agrees with $a(P) \text{star} + b(P) p_1 \in \text{span}\{\text{star}, p_1\}$, while $x = 0$ (a suffix with no edge) is unconstrained; hence $\pi(P) \in \text{span}\{\text{star}, p_1, \delta_0\} = \text{span}\{p_1, q_1, \delta_0\} = U_0$ (the two spans coincide since $\text{star} = p_1 + q_1$). Its three generators are independent on $\{0, 1, 2, 3\}$, and $\pi([1]), \pi([2]), \pi([3])$ – rows 1, 2, 3 of G – are independent since $\det G = 144 \neq 0$, so form a basis. $\square$

Readout (the terminal linear map). For every prefix P the map $x \mapsto \text{Within}(P \cdot [x])$ lies in $V' = \text{span}\{DS(a, \cdot) : a = 0:3\}$. The single-vertex prefix $P = [0]$ gives $DS(0, x) = \text{star}(x + 1)$, a defining

generator of V' . Every other prefix has $|V(P)| \geq 2$ – in particular every one-hub prefix $P = [a]$ with $a \geq 1$, which has $a + 1 \geq 2$ vertices – and there the one-hub gluing $\text{Within}(P \cdot [x]) = a(P) \text{star}(x) + b(P) p_1(x)$ ($x \geq 1$, the $R = [x]$ case of (5.1)) places it in $\text{span}\{p_1, q_1, \delta_0\} \subseteq V'$ (the exceptional edgeless suffix $x = 0$ supplying δ_0 ; e.g. $\delta_0 = (DS(2, \cdot) - 2 DS(1, \cdot))/6$). In particular $DS(a, \cdot) \in V'$ for **all** $a \geq 0$, not merely $a \leq 3$. Hence $\text{Within}(P \cdot [x])$ is the unique element of V' matching the four values $\pi(P)$, which is exactly what the readout $\lambda(x) = G^{-1}(DS(a, x))_a$, $\text{Within} = \sum_d \lambda_d \pi(P)_d$ computes.

Finally the single-vertex prefix $[0]$ – the one exception (it uses the $A = 2$ channel, and (5.4) fails there) – supplies the 4th coordinate: the four profiles $\pi([0]), \dots, \pi([3])$ are the rows of G , a basis of $\mathbb{Q}^4$, so $M(\sigma)$ is uniquely defined by $M(\sigma)\pi([a]) = \pi([a, \sigma])$ for $a = 0, \dots, 3$ – the readout matrix built from the triple values $\text{Within}([a, \sigma, d])$. For $|V(P)| \geq 2$ both sides of $\pi(P \cdot [\sigma]) = M(\sigma)\pi(P)$ are linear functionals of $\pi(P)$ (the left by Proposition 7, the right by definition of $M(\sigma)$) that agree on the basis $\pi([1]), \pi([2]), \pi([3])$ of U_0 , hence on all of $U_0 \ni \pi(P)$; and for $P = [0]$ the identity is the defining relation with $a = 0$. So $\pi(P \cdot [\sigma]) = M(\sigma)\pi(P)$ for every P , and with the readout above the 4-dimensional recurrence computes $\text{Within}(C)$ exactly for every caterpillar. $\square$

Because each interior $M(\sigma)$ maps a genuine profile (a vector of tournament counts) to another, the interior division by 144 is always exact-integer. Against an independent exact counter sharing no logic with the recurrence, the assembled $M(\sigma)$ reproduces Within with zero discrepancies over an extensive battery of caterpillars and profile-collapse tests (Appendix D).

5. Polynomial aggregation

With Within in hand, $T(H)$ is a product over components (§3), and $A_v(t)$ is assembled by a forest recurrence that couples components through the $2^{\binom{c}{2}}$ factor. Writing $w_k(t)$ for the labeled weight of connected caterpillar components on k vertices,

$$w_1(t) = 0, \quad w_k(t) = \sum_{C \text{ caterpillar on } k \geq 2 \text{ vertices}} \frac{k!}{|\text{Aut}(C)|} (-1)^{k-1} \text{Within}(C) I(C)^t \quad (k \geq 2),$$

($w_1 = 0$ because the support graphs H' have no isolated component). The component-coupled recurrence is

$$D_v^{(j)} = \sum_{k=1}^v \binom{v-1}{k-1} w_k(t) 2^j D_{v-k}^{(j+1)}, \quad D_0^{(\cdot)} = 1, \quad A_v^{\text{forest}}(t) = D_v^{(0)},$$

where the depth index j counts already-committed components so that 2^j accrues the $\binom{c}{2}$ between-component edges. The single odd-cycle-sun family contributes a separate additive term $\text{oc}_v(t)$, and $A_v(t) = A_v^{\text{forest}}(t) + \text{oc}_v(t)$; its inclusion–exclusion sign is $(-1)^{e(S)} = (-1)^v$ (a sun on v vertices is connected and unicyclic, $e(S) = v$). Both $I(C)$ and $I(S)$ are evaluated by the standard 2-state spine/cycle recurrence.

Remark. Summing $\text{Within}(C) I(C)^t$ over the $\Theta(2^k)$ caterpillars directly needs none of the machinery below but takes $2^{\Theta(n)}$ operations; this is the variant used for the $L(n)$ table, and it reaches $n \leq 18$.

Polynomial-time aggregation. The enumeration is removable. The obstruction is the factor $I(C)^t$: $I(C)$ is the scalar output of a 2-state spine recurrence, and $I(C)^t$ does not decompose along

the spine. But $I(C)^t = \#\{\text{ordered } t\text{-tuples of independent sets of } C\}$, and a t -tuple is tracked along the spine by the *count* $c \in \{0, \dots, t\}$ of tuple-coordinates whose current spine vertex is “in” – a **symmetric-power** compression of the 2^t raw states to $t+1$. Running this count in lockstep with the Within 4-state transfer, folding the labelled weight $k!/|\text{Aut}(C)|$ into an exponential generating function, and treating the sun term by the analogous cyclic construction, gives a dynamic program that computes $A_v(t)$ – and hence $L(n)$ – with **no enumeration of caterpillars**, in $\text{poly}(n)$ big-integer operations. This establishes:

Theorem L3 (polynomial aggregation). The count-compression dynamic program computes $A_v(t)$, and hence $L(n) = \sum_v \binom{n}{v} 2^{\binom{n-v}{2}} A_v(n-v)$, exactly for every n , in a number of big-integer operations bounded by a fixed polynomial in n .

Proof. We prove the theorem by giving explicit dynamic programs for caterpillars, suns, and forest assembly – the three ingredients being (i) the symmetric count-collapse of $I(C)^t$ to a single size coordinate, (ii) the 2-to-1 directed-spine label encoding reproducing $k!/|\text{Aut}|$ with no palindrome casework, and (iii) the cyclic-seam closure for suns. Each is proved exact by its construction, and their polynomial state dimensions give the bound.

Reduced-word set. A labeled caterpillar on $[k]$ ($k \geq 2$) has m inner (degree- ≥ 2) spine vertices. The cases $m = 0$ (forced to be K_2 , the single edge) and $m = 1$ (a star $K_{1,k-1}$) are separate base cases; for $m \geq 2$ the caterpillar is encoded by its reduced leaf-word, ranging over

$$W_k = \left\{ (\ell_0, \dots, \ell_{m-1}) : m \geq 2, \ell_i \geq 0, \ell_0 \geq 1, \ell_{m-1} \geq 1, \sum_i \ell_i = k - m \right\}.$$

The endpoint conditions $\ell_0, \ell_{m-1} \geq 1$ record that a canonical reduced spine’s endpoints are inner vertices, not leaves, so the spine of a caterpillar is intrinsic. Reading it in its two directions always yields **two distinct** (word, labelling) encodings, because the two spine endpoints carry distinct labels – and this holds *whether or not* the leaf-word is palindromic: a palindromic shape gives the *same* word but with the two endpoint labellings interchanged (still two encodings), a non-palindromic shape gives its two reversed words. Hence the map (encoding) $\rightarrow$ (caterpillar) is uniformly **2-to-1** (the source of the $\frac{1}{2}$ below), and dividing the directed-word sum by 2 reproduces $k!/|\text{Aut}|$ with no separate palindrome correction.

Caterpillar DP for $w_k(t)$. Two ingredients run in lockstep along the spine.

Independent-set count $I(C)^t$. Since $I(C)^t = I(t \cdot C)$ counts spine-patterns weighted by free leaves, and (Lemma (i)) the pattern-count depends on each row only through its “in”-size $c \in \{0, \dots, t\}$, the count compresses to a product of $(t+1) \times (t+1)$ matrices,

$$I(C)^t = \mathbf{1}^\top L_{m-1} P L_{m-2} P \cdots P L_0 e, \quad e_c = \binom{t}{c}, \quad P_{c,c'} = \binom{t-c'}{c}, \quad (L_i)_{c,c} = 2^{\ell_i(t-c)}.$$

Weight Within(C). This is the §4 four-state transfer; with the integer row/matrix data $r(\sigma)[j] = DS(\sigma, j)$, $\tilde{W}(\sigma)[j][d] = \sum_a g_{a,j}(\sigma) \text{adj}(G)[d][a]$, $\tilde{\rho}(\sigma)[d] = \sum_a DS(a, \sigma) \text{adj}(G)[d][a]$ (state $j \in \{0, 1, 2, 3\}$), $\text{Within}(C) = 144^{-(m-1)} \tilde{\rho}(\ell_{m-1}) (\prod_{i=m-2}^1 \tilde{W}(\ell_i) r(\ell_0))$.

Fusion. Fold both, together with the integer label weight $\lambda_i = (k - u_i) \binom{k - u_i - 1}{\ell_i}$ (which telescopes to $k! / \prod_i \ell_i!$, the EGF $1/\ell!$ made integral, u_i the running label count), into an array $V_s[(j, c, u)]$ indexed by transfer-state j , count c , and labels-used u , which the DP advances in **rounds** $s = 0, 1, 2, \dots$. After round s the array holds the fused weight of every reduced prefix with *exactly* $s+1$ spine

hubs; because each of the s completed interior steps multiplied by the integer matrix $\widetilde{W} = 144 M$, every entry of V_s carries the **same** integer factor 144^s . This common, round-uniform power is what makes the read-out division exact and unambiguous – prefixes of different hub-counts are never summed into one array, so no mixed total is ever divided.

- **Round 0 (initialize)** (hub 0, $\ell_0 \geq 1$, each c_0 ; no factor of 144 yet): set $V_0[(j, c_0, 1+\ell_0)] += r(\ell_0)[j] \cdot \binom{t}{c_0} 2^{\ell_0(t-c_0)} \cdot \lambda_0$, with $\lambda_0 = k \binom{k-1}{\ell_0}$.
- **Read-out of round s** (append the terminal hub, forced $\ell = k-1-u \geq 1$; these are the $m = s+2$ -hub caterpillars): accumulate $\tilde{\rho}(\ell)[j'] \cdot \binom{t-c'}{c} 2^{\ell(t-c)} \cdot (k-u) \binom{k-u-1}{\ell} \cdot V_s[(j', c', u)]$ over c, j', c', u , then divide by $144^{s+1} = 144^{m-1}$. This is an exact integer division (Theorem L2) and is unambiguous precisely because every entry of V_s shares the power 144^s .
- **Interior step** $V_s \rightarrow V_{s+1}$ (advance every entry by one hub, any $\ell \geq 0$): set

$$V_{s+1}[(j, c, u+1+\ell)] += \widetilde{W}(\ell)[j][j'] \cdot \binom{t-c'}{c} 2^{\ell(t-c)} \cdot (k-u) \binom{k-u-1}{\ell} \cdot V_s[(j', c', u)],$$

which multiplies the common power by one further 144 (to 144^{s+1}). The new array **replaces** V_s ; this is why each round carries a single hub-count and a single power.

The read-out is taken at every round $s \geq 0$ (each contributing the $m = s+2$ caterpillars) and the results summed. The endpoint constraints $\ell_0, \ell_{m-1} \geq 1$ and total $= k$ restrict the accumulation exactly to $w \in W_k$. Dividing this summed $m \geq 2$ read-out by the 2-to-1 spine reversal and attaching the sign $(-1)^{k-1} = (-1)^{e(C)}$ (a k -vertex tree),

$$w_k(t) = (-1)^{k-1} \left(\underbrace{[k=2] 2 \cdot 3^t}_{K_2} + \underbrace{[k \geq 3] k \text{ star}(k-1) (2^{k-1}+1)^t}_{m=1 \text{ star}} + [k \geq 4] \frac{1}{2} (m \geq 2 \text{ readout total}) \right),$$

the two base cases being the labeled counts of K_2 ($I(K_2) = 3$, one labeling) and of the star $K_{1,k-1}$ (k labelings, Within = $\text{star}(k-1)$, $I = 2^{k-1}+1$).

Sun seam DP for $\text{oc}_v(t)$. The same collapse runs around the cycle $c_0, \dots, c_{g-1}$, but the wrap edge $A_{g-1} \cap A_0 = \emptyset$ breaks the size-only compression. Fix the top set A_0 to a specific set of size c_0 and split every later row by its overlap with A_0 and with A_0^c : track $(a_i, b_i) = (|A_i \cap A_0|, |A_i \setminus A_0|)$. Then the transition is

$$(a', b') \rightarrow (a, b) : \quad \binom{c_0 - a'}{a} \binom{(t - c_0) - b'}{b} \cdot 2^{\ell_i(t-a-b)} \cdot 2^{\binom{\ell_i}{2}},$$

the two binomials being the disjoint one-step extensions inside A_0 and inside A_0^c , the factor $2^{\ell_i(t-a-b)}$ the free leaves, and $2^{\binom{\ell_i}{2}}$ the per-hub Within factor (Lemma L1). Hub 0 initializes the array (for each cycle length g and each size c_0): at state $(a_0, b_0) = (c_0, 0)$ set

$$V_0[(c_0, 0)] = \binom{t}{c_0} 2^{\ell_0(t-c_0)} 2^{\binom{\ell_0}{2}} \cdot v \binom{v-1}{\ell_0}, \quad u = 1 + \ell_0,$$

($\binom{t}{c_0}$ choosing A_0 , the two 2-powers the hub's leaves and Within factor, $v \binom{v-1}{\ell_0}$ the telescoping label weight); hubs $1, \dots, g-1$ apply the transition above (advancing u by $1 + \ell_i$), the last one closed by the seam $a_{g-1} = 0$ (the wrap constraint $A_{g-1} \cap A_0 = \emptyset$). The state (a, b) ranges over $0 \leq a \leq c_0$, $0 \leq b \leq t - c_0$, i.e. $O(t^2)$ states. Reading a labeled sun's cycle from any of its g vertices in either direction gives $2g$ ordered encodings, and because the $g \geq 3$ cycle vertices carry distinct

labels the dihedral group D_g acts freely – so the encoding is exactly **2g-to-1** even for periodic leaf-words. Dividing by $2g$ and attaching the sign $(-1)^v$ (a sun on v vertices has $e(S) = v$ edges),

$$\text{oc}_v(t) = (-1)^v \sum_{\substack{g \text{ odd} \\ 3 \leq g \leq v}} \frac{1}{2g} \sum_{\substack{(\ell_0, \dots, \ell_{g-1}) \\ g + \sum_i \ell_i = v}} \frac{v!}{\prod_i \ell_i!} 2^{1 + \sum_i \binom{\ell_i}{2}} I(S_\ell)^t,$$

the inner double sum evaluated by the seam DP with the same telescoping integer label weight $(v-u) \binom{v-u-1}{\ell}$; the global Within factor 2 pairs with the $2g$ to leave a division by g .

Assembly. With $w_k(t)$ and $\text{oc}_v(t)$ in hand, the forest part is the species convolution

$$D_v^{(j)} = \sum_{k=1}^v \binom{v-1}{k-1} w_k(t) 2^j D_{v-k}^{(j+1)}, \quad D_0^{(j)} = 1, \quad A_v^{\text{forest}}(t) = D_v^{(0)},$$

peeling the block of the current smallest label; the 2^j accrues the j new inter-component edges as a component is added, so the running product $2^0 2^1 \dots 2^{r-1} = 2^{\binom{r}{2}}$. Then

$$A_v(t) = A_v^{\text{forest}}(t) + \text{oc}_v(t), \quad L(n) = \sum_{v=0}^n \binom{n}{v} 2^{\binom{n-v}{2}} A_v(n-v).$$

Complexity (no 2^k appears). Every array is indexed by bounded coordinates: the transfer-state $j \leq 4$, while c, a, b, u, ℓ, k, v each range in $\{0, \dots, n\}$ (as $t = n - v \leq n$). Hence the caterpillar array $V[(j, c, u)]$ has $O(kt)$ entries and $O(kt)$ work per transition ($O(k^3 t^2)$ per w_k), the sun array (a, b, u) has $O(t^2 v)$ entries ($O(v^4 t^5)$ per oc_v), and the forest recurrence has $O(v^2)$ states. Every intermediate entry is an exact integer of $2^{O(n^2)}$ magnitude – hence $O(n^2)$ bits – since it is a signed sum of at most $n!$ label assignments times at most $2^{O(n^2)}$ tournament/independent-set choices; we do *not* claim the tighter $\leq 2^{\binom{n}{2}}$ (individual signed inclusion–exclusion terms may exceed the final count before cancellation, so only the $2^{O(n^2)}$ bit bound is asserted). Summed over $v \leq n$, $t = n - v$, the whole computation is a fixed polynomial number of big-integer operations (a loose $O(n^{10})$; the measured scaling is in Appendix D). Since every operand has $O(n^2)$ bits, each such operation costs $\text{poly}(n)$ bit-operations, so the algorithm runs in **polynomial bit complexity** – with **no enumeration of caterpillars or suns** and no 2^k state anywhere. This is the assertion of Theorem L3. ■

An implementation reproduces $L(7), \dots, L(18)$ exactly, with measured running time scaling as a fixed polynomial ($\approx n^{7.3}$) against two exponential competitors (Appendix D); we thus have a **proved polynomial-time algorithm** for $L(n)$. The Burnside/cycle-index step yielding the count of $\gamma \geq 3$ tournaments up to isomorphism is a separate small- n layer, developed in Appendix D.

6. Results and validation

The method of §§1–5 is proved, so every value below is a theorem; the corroboration notes record what independent cross-check is additionally available. The $A_v(t)$ were cross-checked term by term for all $v \leq 10$ against an *independent* enumeration of the contributing structures (caterpillar forests and odd-cycle suns), each weighted by a Within value obtained from its defining count rather than from the recurrence: exact agreement. Direct brute-force enumeration of all $2^{\binom{n}{2}}$ tournaments

confirms $L(7), L(8), L(9)$; the four-state Within recurrence matches exact enumeration with zero mismatches on an extensive test battery (Appendix D).

n	$L(n) = \#\{\gamma \geq 3\}$ (labeled)	status
7	240	direct enumeration (2^{21}) + full Burnside
8	109440	direct enumeration (2^{28}) + full Burnside
9	77274880	direct enumeration (2^{36})
10	107731095040	Burnside-corroborated vs. independent iso(10)
11	277644146489600	new; also passes a Burnside congruence (App. D)
12	1259251974345052160	new; also passes a Burnside congruence (App. D)
13	10059739307720354796544	new
14	144025884971245392244400128	new
15	3764299177453034811662226208768	new
16	182456135219244423561589552542810112	new
17	16611757357122373923247804720005425659904	new
18	2870353532067896148226715034553628871332397056	new

Here “direct enumeration” means examining all $2^{\binom{n}{2}}$ tournaments; we ran this for $n = 7, 8, 9$ (up to 2^{36} , sharded). $L(10)$ was corroborated by an independent Burnside assembly against iso(10); $L(11), L(12)$ pass Burnside congruences (Appendix D). All entries are theorems; the corroboration column records only what independent cross-check is available.

As a concrete payoff, the density $L(n)/2^{\binom{n}{2}}$ – the probability that a uniformly random labeled tournament has $\gamma \geq 3$ – can be read off directly: it rises from 0.011% at $n = 7$ through 0.31% at $n = 10$ and 3.3% at $n = 13$ to 25.1% at $n = 18$, exhibiting the threshold climb concretely rather than through asymptotic bounds.

The polynomial algorithm is not confined to this window; we ran it through $n = 30$, where it evaluates the aggregate $L(n) = \#\{\gamma \geq 3\}$ (which for $n \geq 19$ exceeds $\#\{\gamma = 3\}$, per the scope note of §1). For instance

$$L(19) = 949209776866692815063198172251754624787427891544064,$$

$$L(20) = 604915380532280302020187569457614900100873131394954166272.$$

We do not tabulate $n \geq 19$ further: these no longer isolate $\gamma = 3$, and separating the $\gamma = 3$ and $\gamma \geq 4$ parts would require a structure theorem for dominating triples that we do not have.

(For $n \leq 18$ every tournament has $\gamma \leq 3$, so $L(n) = \#\{\gamma = 3\}$ throughout.) The full domination distribution (labeled, $\gamma \in \{1, 2, 3\}$, with $\gamma_1 = n 2^{\binom{n-1}{2}}$ and $\gamma_2 = 2^{\binom{n}{2}} - \gamma_1 - \gamma_3$; each row sums to $2^{\binom{n}{2}}$) is, for $n = 2, \dots, 14$ – the two-paradox class $\gamma = 3$ first appearing at $n = 7$, the smallest order admitting a tournament with $\gamma \geq 3$:

n	$\gamma = 1$	$\gamma = 2$	$\gamma = 3$
2	2	0	0
3	6	2	0
4	32	32	0
5	320	704	0
6	6144	26624	0
7	229376	1867536	240
8	16777216	251548800	109440
9	2415919104	66226282752	77274880
10	687194767360	34389446226432	107731095040
11	387028092977152	35364124779497216	277644146489600
12	432345564227567616	72095378756265586688	1259251974345052160
13	959230691832896684032	291212484904104042195968	10059739307720354796544
14	4231240368651202111471616	2327622953230863955442376704	144025884971245392244400128

Isomorphism classes. The number of *isomorphism classes* of two-paradox tournaments follows from the labeled counts by Burnside’s lemma over S_n (the identity term is $L(n)$; Appendix D):

n	7	8	9	10	11	12
$\text{iso}(n)$	1	5	226	29816	6959159	2629321652

Here $\text{iso}(7), \dots, \text{iso}(10)$ agree with the complete tournament catalogues through order 10; $\text{iso}(11)$ and $\text{iso}(12)$ come from an independent isomorph-free generation (checksummed against the tournament sequence A000568) and satisfy the Burnside congruences derived in Appendix D. The unique class at $n = 7$ is the Paley (quadratic- residue) tournament QR_7 . The Burnside congruence checks and detailed test records are collected in Appendix D.

Appendix A. Channel formulas and finite certificate

Write $p_A(\ell) = 2^{\binom{\ell}{2}} A^\ell$ and $q_A(\ell) = \ell 2^{\binom{\ell-1}{2}} A^{\ell-1}$ for $A \in \{1, 2, 4\}$ (the $A = 8$ slot is unused). The single-vertex value is $\text{star}(m) = p_1(m) + q_1(m)$.

The matrix G ($G[a][d] = DS(a, d) = \text{Within}([a, d])$, symmetric, $\det G = 144$):

$$G = \begin{pmatrix} 2 & 4 & 14 & 96 \\ 4 & 4 & 8 & 30 \\ 14 & 8 & 16 & 60 \\ 96 & 30 & 60 & 224 \end{pmatrix}, \quad \text{adj}(G) = \begin{pmatrix} 0 & -48 & 24 & 0 \\ -48 & -30112 & 22928 & -2088 \\ 24 & 22928 & -17416 & 1584 \\ 0 & -2088 & 1584 & -144 \end{pmatrix} = 144 G^{-1}.$$

Double-star channels $DS(L, d) = \text{Within}([L, d])$ (for $L \geq 1$; $DS(L, 0) = \text{star}(L + 1)$):

$$DS(L, 1) = 3p_1(L) + q_1(L), \quad DS(L, 2) = 6p_1(L) + 2q_1(L), \quad DS(L, 3) = 22p_1(L) + 8q_1(L).$$

Triple channels $g_{a,c}(\ell) = \text{Within}([a, \ell, c])$, $0 \leq a \leq c \leq 3$ (the ten cases; $g_{c,a} = g_{a,c}$):

$$\begin{aligned} g_{0,0} &= 2p_2 + 2p_4 + 2q_4, & g_{0,1} &= p_1 + 3p_2 + q_2, & g_{0,2} &= 2p_1 + 6p_2 + 2q_2, \\ g_{0,3} &= 8p_1 + 22p_2 + 8q_2, & g_{1,1} &= 4p_1, & g_{1,2} &= 8p_1, \\ g_{1,3} &= 30p_1, & g_{2,2} &= 16p_1, & g_{2,3} &= 60p_1, \\ g_{3,3} &= 224p_1. \end{aligned}$$

Finite certificate (derivation of the channels).

The single-vertex value, by direct counting. The independent sets of the star $K_{1,m}$ are the leaf-subsets and $\{\text{centre}\}$, and validity means every vertex beating the centre loses to all leaves. At most one leaf beats the centre: if two leaves u, u' both beat it, then whichever of $u \rightarrow u'$, $u' \rightarrow u$ holds makes that leaf a common in-neighbour of the other's centre-edge, which the reformulation forbids. If none, the centre beats all m leaves and the $\binom{m}{2}$ leaf-edges are free ($2^{\binom{m}{2}}$ completions); if exactly one (m choices of the special leaf u^*), u^* loses to the other $m-1$ leaves whose $\binom{m-1}{2}$ edges are free ($m 2^{\binom{m-1}{2}}$). Hence $\text{star}(m) = 2^{\binom{m}{2}} + m 2^{\binom{m-1}{2}} = p_1(m) + q_1(m)$.

The double-star and triple channels, by core-config enumeration. Fix the varying hub s of the channel and apply the Leaf-Factor Lemma: each internally-valid orientation τ of the finite core (everything not incident to the leaves of s) contributes $f_\tau(\ell) = p_{A_\tau}(\ell) + A_\tau^* q_{A_\tau}(\ell)$, and the channel is $\sum_\tau f_\tau(\ell)$. Enumerating the finitely many τ and recording, for each realized value A , the number of configs with that A (the coefficient on p_A) and the sum of their A^* values (the coefficient on q_A) fixes the two coefficients of each channel. The core of a channel is itself a bounded caterpillar fragment – for the triple $g_{a,c} = \text{Within}([a, \ell, c])$ it is the three-hub caterpillar $[a, \cdot, c]$ with $a, c \leq 3$, at most $3 + a + c \leq 9$ vertices, hence at most 2^9 orientations to inspect – so the enumeration is finite. All fourteen double-/triple-channel cases are recorded below in aggregate (the right column is the resulting closed form, whose p_A -coefficient is the A -count and whose q_A -coefficient is the A^* -sum, so the table and Appendix A determine each other):

channel	(A : count)	(A : $\sum A^*$)	closed form (App. A)
$DS(L, 0)$	$\{1:1, 2:1\}$	$\{2:1\}$	$\text{star}(L+1)$
$DS(L, 1)$	$\{1:3\}$	$\{1:1\}$	$3p_1 + q_1$
$DS(L, 2)$	$\{1:6\}$	$\{1:2\}$	$6p_1 + 2q_1$
$DS(L, 3)$	$\{1:22\}$	$\{1:8\}$	$22p_1 + 8q_1$
$g_{0,0}$	$\{2:2, 4:2\}$	$\{4:2\}$	$2p_2 + 2p_4 + 2q_4$
$g_{0,1}$	$\{1:1, 2:3\}$	$\{2:1\}$	$p_1 + 3p_2 + q_2$
$g_{0,2}$	$\{1:2, 2:6\}$	$\{2:2\}$	$2p_1 + 6p_2 + 2q_2$
$g_{0,3}$	$\{1:8, 2:22\}$	$\{2:8\}$	$8p_1 + 22p_2 + 8q_2$
$g_{1,1}$	$\{1:4\}$	$\emptyset$	$4p_1$
$g_{1,2}$	$\{1:8\}$	$\emptyset$	$8p_1$
$g_{1,3}$	$\{1:30\}$	$\emptyset$	$30p_1$
$g_{2,2}$	$\{1:16\}$	$\emptyset$	$16p_1$
$g_{2,3}$	$\{1:60\}$	$\emptyset$	$60p_1$
$g_{3,3}$	$\{1:224\}$	$\emptyset$	$224p_1$

(The DS rows also carry an $A=0$ boundary config contributing 0; $A=0$ terms never affect p_A, q_A .)

Exhaustiveness. Since the Leaf-Factor Lemma pins each channel's ℓ -dependence to the fixed two-dimensional block $\{p_A, q_A\}$, and the core is a bounded fragment with only finitely many orientations, the complete enumeration above determines the two coefficients – and hence the whole channel function – for **all** ℓ . This is a proof, not a fit: no interpolation through sampled values is used; the ℓ -shape is fixed a priori and only its two coefficients are read off the finite enumeration. (The table records, for each channel, the aggregate A -count and A^* -sum over the core configurations rather

than the individual configurations; since the core has at most $2^{\binom{9}{2}}$ orientations, a reader may in principle regenerate the aggregates directly.)

Transfer convention. State $R \in \mathbb{Z}^4$, $R[j]$ = the value for the processed prefix followed by a spine vertex carrying j leaves. Initialize $R[j] = DS(\ell_0, j)$. For each interior leaf-count ℓ (in spine order $\ell_1, \dots, \ell_{m-2}$): set $S[a] = \sum_d \text{adj}(G)[d][a] R[d]$, then $R'[j] = \frac{1}{144} \sum_a g_{a,j}(\ell) S[a]$ (an exact integer division, by Theorem L2, so $R \in \mathbb{Z}^4$ throughout). Read out with the last symbol $x = \ell_{m-1}$: $\text{Within} = \frac{1}{144} \sum_d (\sum_a DS(a, x) \text{adj}(G)[d][a] R[d])$. Base cases: $m = 1$ gives $\text{star}(\ell_0)$; $m = 2$ is obtained by the *same* readout – initialize $R[j] = DS(\ell_0, j)$ and read out at $x = \ell_1$ – which supplies double stars $DS(\ell_0, \ell_1)$ for arbitrary ℓ_0, ℓ_1 (e.g. $\text{Within}([4, 4]) = 12288$), beyond the $d \leq 3$ closed forms tabulated above. The convention reproduces every channel closed form and the values $L(7), \dots, L(18)$ exactly.

Appendix B. Odd-cycle-sun orientation proof

This appendix proves the four claims (a)–(d) of Lemma L1 (§3). Write the cycle as $c_0, \dots, c_{g-1}$ (indices mod g), with ℓ_i leaves on hub c_i , and work inside any valid tournament, using only the dominating-pair condition on the cycle edges and H -independence of out-neighbourhoods.

(a) Exactly two orientations of the cycle-and-chord tournament. *No local sink:* if $c_{i-1} \rightarrow c_i \leftarrow c_{i+1}$, then domination of $\{c_{i-1}, c_i\}$ (third vertex c_{i+1}) forces $c_{i-1} \rightarrow c_{i+1}$ and domination of $\{c_i, c_{i+1}\}$ (third vertex c_{i-1}) forces $c_{i+1} \rightarrow c_{i-1}$ – a contradiction. In the cyclic F/B word of the cycle-edge orientations, $\#\text{sinks} = \#\text{sources}$; with no sink there is no source, hence no transition, so the word is constant: the cycle is a **directed cycle** (forward or backward – two choices). Fixing $c_i \rightarrow c_{i+1}$, induction on the gap d forces every chord $\{c_i, c_{i+d}\}$: when d is odd $c_{i+d} \in N^+(c_i)$, and H -independence forbids the cycle-adjacent c_{i+d+1} , forcing $c_{i+d+1} \rightarrow c_i$ (gap $d+1$ even, inward); when d is even, domination of $\{c_i, c_{i+1}\}$ rules out the reverse, forcing $c_i \rightarrow c_{i+d+1}$ (gap $d+1$ odd, outward). So $c_i \rightarrow c_{i+d}$ iff d is odd. Gap d from c_i and gap $g-d$ from c_{i+d} name the same edge, and the two forcings agree **iff g is odd**; hence odd g yields exactly the two rotational tournaments R (each valid) and even g yields 0 (the wrap edge $\{c_0, c_{g-1}\}$ is forced both ways). For each hub c_i , $N^+(c_i) \cap \text{hubs}$ and $N^-(c_i) \cap \text{hubs}$ are maximum independent sets of C_g , and c_i beats exactly one cycle-neighbour.

(b) Each leaf's orientation is forced against every foreign hub. No leaf beats its own hub: if a leaf $u \rightarrow c_i$, the cycle-neighbour $c_i^* \rightarrow c_i$ is forced (independence of $N^+(u) \ni c_i$) to satisfy $c_i^* \rightarrow u$, making c_i^* a common in-neighbour of the leaf edge $\{u, c_i\}$ – contradiction; so $c_i \rightarrow u$. Domination of $\{u, c_i\}$ then forces $N^+(u) \supseteq N^-(c_i) \cap \text{hubs}$ (a maximum independent set), and independence caps it, so $N^+(u) \cap \text{hubs} = N^-(c_i) \cap \text{hubs} = \{c_a : (i-a) \text{ odd}\}$ is **forced**; in particular every leaf-foreign-hub edge is determined by the parity of the offset.

(c) Forced orientation between leaves at different hubs. For a leaf u of c_i and a leaf v of c_j ($j \neq i$), exactly one of $(j-i), (i-j)$ is odd. If $(j-i)$ is odd then $v \rightarrow c_i$ by (b), so $v \rightarrow u$ would put both u and c_i into $N^+(v)$, i.e. the sun edge $\{u, c_i\}$ inside $N^+(v)$ – impossible; hence $u \rightarrow v$. The symmetric case forces $v \rightarrow u$. So every foreign leaf-leaf edge is **forced**.

(d) Freedom only within same-hub leaf sets. Two leaves u, u' of the same hub c_i are not joined by any H -edge (their only H -neighbour is c_i), so the pair $\{u, u'\}$ carries no domination constraint of its own; and since $c_i \in N^+(u) \cap N^+(u')$ is impossible to violate here ($c_i \notin N^+(u), N^+(u')$ as

c_i beats each leaf by (b)), orienting $\{u, u'\}$ either way adds no H -edge to any out-neighbourhood. Hence both orientations are valid and independent. The $\binom{\ell_i}{2}$ intra-hub edges are thus free and independent across hubs, and any such completion is valid (each out-neighbourhood is checked H -independent). Each of the two rotational orientations therefore contributes $\prod_i 2^{\binom{\ell_i}{2}}$, giving $\text{Within} = 2 \cdot \prod_i 2^{\binom{\ell_i}{2}} = 2^{1+\sum_i \binom{\ell_i}{2}}$. $\square$ Verified exhaustively: $C_3, \dots, C_7$, all suns on ≤ 8 vertices, and each zero-case family, by independent enumeration.

Appendix C. Worked example: $n = 7$

Master sum. With $A_v(t) = \sum_H (-1)^{e(H)} T(H) I(H)^t$ over labeled minimum-degree- ≥ 1 graphs on v vertices:

- $v = 2$: the only such H is the edge K_2 ; $e = 1$, $T(K_2) = 2$, $I(K_2) = 3$ (namely $\emptyset, \{a\}, \{b\}$). So $A_2(t) = -2 \cdot 3^t$, and $A_2(5) = -486$.
- $v = 3$: the labeled graphs are the 3 paths P_3 and the 1 triangle K_3 . For P_3 : $e = 2$, $T = \text{Within}(P_3) = 4$, $I = 5$. For K_3 (an odd-cycle sun with no leaves): $e = 3$, $T = \text{Within} = 2$, $I = 4$. Thus $A_3(t) = 3 \cdot 4 \cdot 5^t - 2 \cdot 4^t = 12 \cdot 5^t - 2 \cdot 4^t$, and $A_3(4) = 6988$.

Continuing (the higher A_v are assembled the same way from stars, double-stars, paths and suns), the master formula $L(7) = \sum_v \binom{7}{v} 2^{\binom{7-v}{2}} A_v(7-v)$ gives:

v	$\binom{7}{v}$	$2^{\binom{7-v}{2}}$	$A_v(7-v)$	term
0	1	2097152	1	+2097152
2	21	1024	-486	-10450944
3	35	64	6988	+15653120
4	35	8	-39672	-11108160
5	21	2	124896	+5245632
6	7	1	-242352	-1696464
7	1	1	259904	+259904

($v = 1$ contributes 0.) The terms sum to

$$L(7) = 2097152 - 10450944 + 15653120 - 11108160 + 5245632 - 1696464 + 259904 = \boxed{240}.$$

Isomorphism count. The odd cycle types of 7 are $[7], [5, 1, 1], [3, 3, 1], [3, 1^4], [1^7]$, with edge-orbit counts $E = 3, 5, 7, 11, 21$ and coefficients $7!/z_\lambda = 720, 504, 280, 70, 1$. The S_2 -invariant counts are $S([7]) = 2$, $S([5, 1, 1]) = 0$, $S([3, 3, 1]) = 12$, $S([3, 1^4]) = 0$, and $S([1^7]) = L(7) = 240$. Hence

$$\text{iso}(7) = \frac{1}{5040} (720 \cdot 2 + 280 \cdot 12 + 1 \cdot 240) = \frac{1440 + 3360 + 240}{5040} = \frac{5040}{5040} = 1.$$

The single class is the Paley (quadratic-residue) tournament QR_7 , with $|\text{Aut}| = 21$; its labeled orbit has size $7!/21 = 240$, matching $L(7)$ exactly.

Appendix D. Computational records, timing, and Burnside checks

Timing. We have implemented this; it reproduces $L(7), \dots, L(18)$ exactly, and its measured running time scales as a fixed polynomial. Figure 1 plots the measured single-threaded wall-clock time against n on a semilogarithmic axis, alongside the two exponential competitors. A least-squares fit to the measured points ($7 \leq n \leq 32$) gives $\approx C n^{7.3}$ for the polynomial algorithm, against $\approx 2^{1.10n}$ for the same signed sum evaluated by naïve term-by-term enumeration, and $2^{\binom{n}{2}}$ for brute-force evaluation of every tournament. On these axes an exponential method is a straight line; the polynomial method’s downward curvature away from the enumeration line is the visible signature of its sub-exponential growth, and it is the only one of the three that remains on a human time-scale as n grows. This is independent numerical corroboration of Theorem L3, not a substitute for it. We therefore have a **proved polynomial-time algorithm** for $L(n)$.

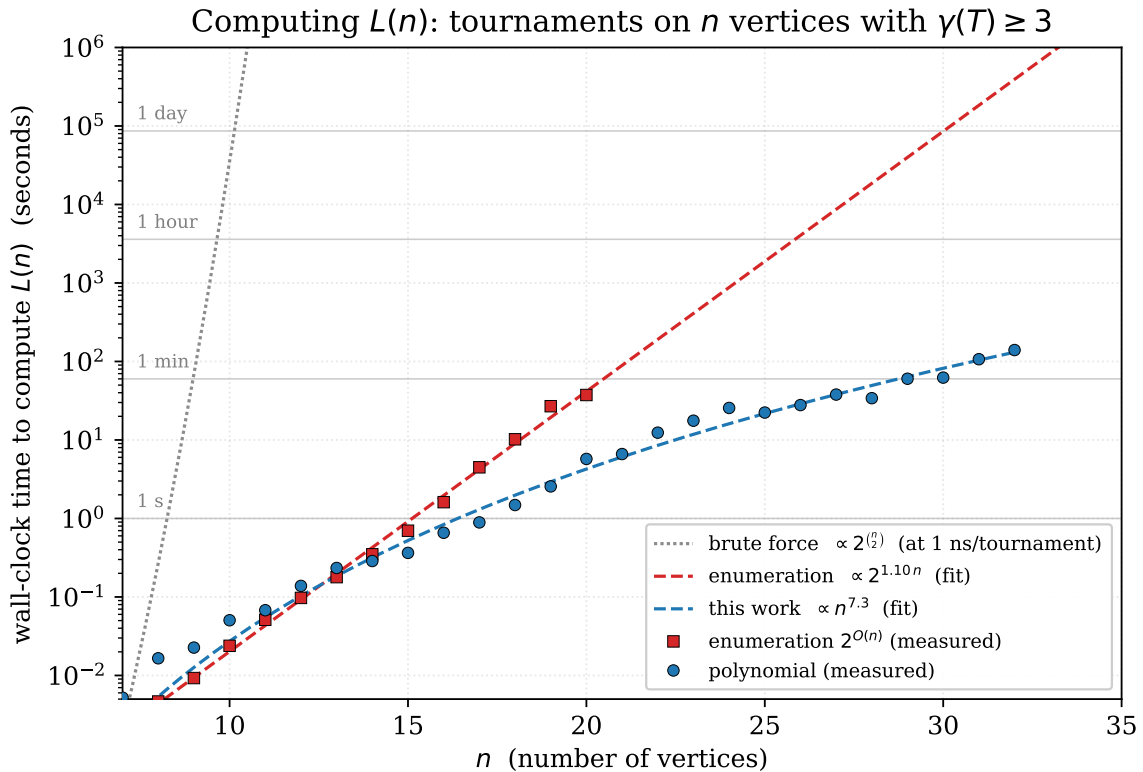


Figure 1: Wall-clock time to compute $L(n)$ versus n (logarithmic time axis), with asymptotic comparison slopes superimposed. Blue: the polynomial algorithm of Theorem L3 (measured points; dashed least-squares fit $\propto n^{7.3}$). Red: the same signed master sum evaluated by naïve term-by-term enumeration of caterpillars (measured points; fit $\propto 2^{1.10n}$). Grey: the $2^{\binom{n}{2}}$ brute force over all tournaments, charged an optimistic 1 ns per tournament, which leaves the frame by $n = 10$. On a semilogarithmic axis an exponential running time is a straight line, so the polynomial curve’s bend below the enumeration line directly exhibits the separation proved in Theorem L3. Horizontal guides mark human time-scales. All timings are single-threaded exact big-integer arithmetic.

Within closed forms (verified by exhaustive enumeration over all orientations): star $K_{1,m} = 2, 4, 14, 96, 1344$; paths $= 2, 4, 4, 4$; double stars $DS(a, 1) = 4, 8, 30, 224, 3392$, $DS(3, 3) = 224$,

$DS(4, 3) = 1664$, $DS(5, 2) = 6784$. The four-state recurrence reproduces Within with **zero mismatches** on all 44 non-isomorphic caterpillars through order 8 (against exact enumeration) and, against an independent pruned counter, on 241 bounded-leaf (≤ 3) spine-encodings through order 9 and 66 targeted cases to $k \approx 13$ (§4).

Independent verification. The crux (Lemma 5’s “two independent covers = the two colour classes”) is confirmed exactly on connected bipartite graphs; and, against an independent exact counter sharing no logic with the recurrence, the assembled $M(\sigma)$ reproduces Within with **zero discrepancies** over all 44 caterpillars through order 8, 241 bounded-leaf encodings through order 9, 66 targeted cases to $k \approx 13$, and a battery of 1131 profile-collapse tests (spines to length 8, including the exceptional [0]).

Isomorphism-class counts. Applying Burnside/cycle-index over the symmetric group,

$$\text{iso}(n) = \frac{1}{n!} \sum_{\lambda \text{ all-odd}} \frac{n!}{z_{\lambda}} S(\lambda),$$

where $S(\lambda)$ is the number of π -invariant S_2 tournaments for a permutation of cycle type λ (only odd cycle types can fix a tournament). The identity term is $S(1^n) = L(n)$ from the master formula of §2; each non-identity term is computed by direct orientation enumeration over edge-orbits (Gray-code, $O(2^{E(\lambda)})$). *Many* cycle types have far fewer edge orbits than $\binom{n}{2}$ and are cheap; but the **near-identity** types (a small cycle plus many fixed points) have $E(\lambda) = \Theta(n^2)$ – e.g. $E([3, 1^{n-3}]) = \frac{1}{2}(n^2 - 5n + 8)$ – so this Burnside layer is *not* single-exponential and is a separate small- n device (this appendix), not part of the $2^{O(n)}$ labeled-count method.

Isomorphism classes. For $n \leq 10$ the Burnside sum was assembled with *all* cycle-type terms computed exactly (each $S(\lambda)$ by Gray-code orbit enumeration, the identity term by the pipeline), giving exact integer quotients

$$\text{iso}(7, \dots, 10) = 1, 5, 226, 29816.$$

These agree with direct enumeration of all non-isomorphic tournaments (McKay’s tournament catalogues, complete through order 10).

For $n = 11, 12$ the values $\text{iso}(11) = 6959159$ and $\text{iso}(12) = 2629321652$ are **computed here by isomorph-free generation**, an algorithm independent of the transfer-matrix pipeline; McKay’s general catalogues stop at order 10 and are not a source for them. Since S_2 is *not* hereditary under vertex deletion (QR_7 is S_2 but every tournament on < 7 vertices has $\gamma \leq 2$), the generator does **not** prune non- S_2 intermediates: canonical augmentation builds all n -vertex tournaments up to isomorphism – each order’s total checksummed against the tournament sequence A000568 (orders 9, 10, 11, 12: 191536, 9733056, 903753248, 154108311168) – and the S_2 property is applied as a *post-filter* on completed tournaments. Generating and filtering all 154,108,311,168 order-12 tournaments is a substantial computation; here $\text{iso}(11), \text{iso}(12)$ enter only as an independent consistency cross-check, not as part of the labeled-count method. The Burnside sum is used only as a *consistency check* linking the two independent computations. Concretely, for $n = 11$ let

$$K = \sum_{\substack{\lambda \text{ odd}, \lambda \neq 1^{11} \\ \lambda \neq [3, 1^8]}} \frac{11!}{z_{\lambda}} S(\lambda) = 32,677,258,240$$

be the sum of all non-identity cycle-type terms except $[3, 1^8]$ (each $S(\lambda)$ by Gray-code enumeration; $E(\lambda) \leq 30$, all brute-feasible). Since $11!/z_{[3, 1^8]} = 330$, the Burnside identity gives

$$\text{iso}(11) \cdot 11! - L(11) - K = 330 \cdot S([3, 1^8]) \equiv 0 \pmod{330},$$

and indeed $\text{iso}(11) \cdot 11! - L(11) - K = 110,534,223,360 = 330 \cdot 334,952,192$, so the transfer-matrix $L(11)$ is consistent with the independently-generated $\text{iso}(11)$ and forces the clean non-negative integer $S([3, 1^8]) = 334952192$. (Note $\text{iso}(11) \cdot 11! - L(11)$ *without* K is $\equiv 220 \not\equiv 0 \pmod{330}$; the K term is essential.) The analogous check for $n = 12$: with $K_{12} = \sum_{\lambda \neq 1^{12}, [3, 1^9]} (12!/z_\lambda) S(\lambda) = 6,066,211,092,480$ and $12!/z_{[3, 1^9]} = 440$,

$$\text{iso}(12) \cdot 12! - L(12) - K_{12} = 191,237,666,498,560 = 440 \cdot 434,631,060,224,$$

forcing $S([3, 1^9]) = 434631060224$. These are consistency checks, not independent recomputations of the iso counts.

We do **not** report $\text{iso}(13)$ or $\text{iso}(14)$: they require the near-identity $S(\lambda)$ terms (e.g. $[3, 1^{10}]$, $E = 56$), whose brute cost is infeasible and whose method computation was not completed. Note that these near-identity Burnside terms cost $2^{\Theta(n^2)}$ (for $\lambda = [3, 1^{n-3}]$, $E(\lambda) = \frac{1}{2}(n^2 - 5n + 8)$), so the isomorphism layer is a **separate small- n validation layer**, not part of the $2^{O(n)}$ labeled-count method of §§1–6.